

S. V. National Institute of Technology (SVNIT), Surat-07, Gujarat, India

Department of Electronics Engineering

Mid Semester Examination (September-2023)

B. Tech - III, 5th Sem

EC305 - Digital Signal Processing

Total Duration : 90 Minutes

Total Marks : 38

Max Allowed Marks : 30

Instruction: (01) Attempt all questions. & Maintain proper sequence.
 (02) Assume necessary Data if required.
 (03) Figure to the right indicates contains mark.

1. (a) Determine whether or not, each of the following signal is periodic. In case as signal is periodic, specify its fundamental period. Specify whether following signal is energy signal or power signal too. (04)

(1) $x(n) = \cos(\pi + 0.2n)$.

(2) $x(n) = e^{j\frac{\pi}{16}n} \cos(n\frac{\pi}{17})$.

Attempt Any TWO from following questions (b) to d).

- (b) Determine the response of system by applying convolution method. The system's impulse response is as; (03)

$$h(n) = \left(\frac{1}{3}\right)^n u(n-3),$$

and the input signal as; $x(n) = \left(\frac{1}{6}\right)^{n-6} u(n)$

- (c) The frequency response of the Linear-Shift-Invariant system is given by: (03)

$$H(e^{j\omega}) = \frac{1 - \frac{1}{4}e^{-j\omega}}{1 + \frac{1}{2}e^{-j\omega}}$$

Find out the frequency response of its inverse system. And find unit step response of such inverse system.

- (d) Consider the a system described by the difference equation: (03)

$$y(n) = y(n-1) - y(n-2) + 0.5x(n) + 0.5x(n-1)$$

Find the response to the system if the input $x(n) = (0.5)^n u(n)$ using Homogeneous and Particular solution methodology.

2. (a) The causal LTI system is described by the difference equation : (02)

$$2y(n) = \alpha y(n-2) - 2x(n) + \beta x(n-1)$$

What should be the value of α and β for the system to be stable.

- (b) A LTI system is characterized by the system function: (03)

$$H(z) = \frac{3 - 4z^{-1}}{1 - z^{-1} + 0.5z^{-2}}$$

Specify Region of Convergence (RoC), and find the impulse response $h(n)$ if.

- (1). The system is stable.
 (2). The system is causal.
 (3). The system is anti-causal.

$2 + 3e^{-j\pi k} - 2e^{-j\pi k}$
 $2 - 3j + 2$

$12 \sqrt{1-4}$
 2

$12 \sqrt{1-4}$

(c) The sequence $y(n]$ is the solution to the LCCoDE, with $x(n) = \delta(n)$.

(03)

$$y(n) - \frac{3}{2}y(n-1) + \frac{1}{2}y(n-2) = x(n), \quad n \geq 0$$

Find a set of initial conditions on $y(n]$ for $n < 0$, so that $y(n) = 1$ for $n > 0$

3. (a) Consider the filter,

$$y(n) = 0.9y(n-1) + bx(n)$$

Determine b so that $|H(0)| = 1$, and also determine the frequency at which $|H(\omega)| = 1/\sqrt{2}$

- OR -

(a) Consider a filter implemented using the difference equation ,

(03)

$$y(n) = \frac{1}{4}x(n) + \frac{1}{2}x(n-1) + \frac{1}{4}x(n-2)$$

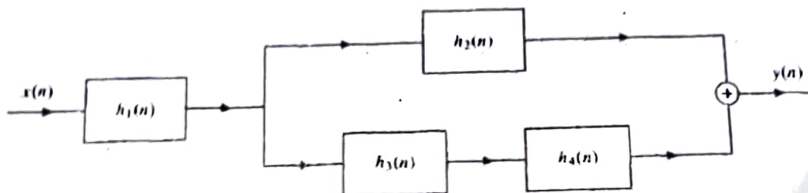
i. What is the DC gain of the filter.

ii. Sketch the pole-zeros diagram of this function.

iii. What is behavior of this filter, e.g. LPF, HPF etc..

(b) Consider an interconnected LTI system shown in figure.

(03)



(i) Find out overall frequency response of the system if the frequency response is defined as $H_1(e^{j\omega})$, $H_2(e^{j\omega})$, $H_3(e^{j\omega})$, $H_4(e^{j\omega})$.

(ii) Find the frequency response of the system if the impulse response of the system is defined as, $h_1(n) = \delta(n) + 2\delta(n-2) + \delta(n-4)$, $h_2(n) = h_3(n) = (0.2)^n u(n)$ and $h_4(n) = \delta(n-2)$.

4. (a) Find the 10-point DFT of the sequence.

$$x(n) = \cos\left(\frac{3\pi}{5}n\right) \cdot \cos\left(\frac{4\pi}{5}n\right)$$

- OR -

(a) if $X(k)$ is the 10-point DFT of the sequence $x(n) = \delta(n-1) + 2\delta(n-4) - \delta(n-7)$. What is the sequence, $y(n)$, that has 10-point DFT is given by,

$$Y(k) = 2X(k) \cos\left(\frac{6\pi k}{N}\right)$$

(b) Suppose that we would like to find the N -point DFT sequence, where $N = 3^m$; N is power of 3. Developed a radix-3 decimation-in-time FFT algorithm, and Draw the corresponding flow-graph for $N = 9$. How many multiplication are required for this FFT algorithm. Can the computation be performed out-of-place?

♣ End of Pages ♣

Question:	1	2	3	4	Total
Points:	10	8	6	6	30